MINI PROJECT REPORT

PLC-BASED AUTOMATION SYSTEM DESIGN

Automatic Conveyor Sorting and Counting System

Project Area: Robotics / Industrial Automation **Controller:** PLC

ABSTRACT

This mini project proposes a PLC-based automation system for detecting, transporting, sorting and counting packages on a conveyor. The system combines digital sensors, push buttons, a programmable logic controller, motor-control outputs and an optional sorting actuator. The PLC executes a cyclic program: it reads field inputs, evaluates the sequence and safety/permissive conditions, updates outputs, and then repeats the scan. The project demonstrates digital I/O, latching, timers, counters, interlocking, fault handling and operator indication. The architecture is intentionally modular so that an educational prototype can later be expanded with HMI, VFD, industrial networking, machine vision and production data collection.

Keywords: PLC, conveyor, ladder logic, sensors, automation, HMI, Industry 4.0

1. INTRODUCTION

A programmable logic controller is an industrial computer designed to monitor inputs and control outputs according to a user program. PLCs are valued for deterministic control, rugged operation and the ability to modify machine behavior through software. IEC 61131-3:2025 defines the syntax and semantics of PLC programming languages, including Ladder Diagram (LD), Function Block Diagram (FBD), Structured Text (ST) and Sequential Function Chart (SFC). [1]

The proposed application is a small conveyor line. A package enters the conveyor, sensors identify its presence and position, the PLC controls the motor, and an end sensor confirms completion. An optional diverter can send selected packages to a reject or alternate lane. The design therefore illustrates the fundamental automation loop: *sense* → *decide* → *actuate* → *verify*.

2. PROJECT OBJECTIVES

The objectives are to design a practical PLC control architecture; define digital inputs and outputs; create a sequential control strategy; demonstrate motor start/stop, sensor detection, timing and counting; include basic fault and safety interlocks; and provide a scalable platform for future HMI and Industry 4.0 features.

3. SYSTEM DESCRIPTION

The system uses a conveyor belt driven by an AC motor. For an educational prototype, a low-voltage motor or suitably isolated motor driver may be used. In an industrial implementation, the PLC output would normally command a properly rated contactor, starter or VFD rather than directly switching a motor. Sensors are positioned along the conveyor to detect the incoming package, confirm its process position and identify arrival at the end station.

The PLC receives these discrete signals and executes the programmed sequence. A Start button places the system in automatic mode, while Stop ends normal operation. A Reset input clears a completed-item counter or acknowledged fault. A diverter output can be used when a package must be sorted according to a predefined condition.

4. COMPONENTS AND FUNCTIONS

Component	Purpose
PLC CPU	Runs the control program and manages I/O.
24 V DC supply	Provides control power for PLC/sensors.
Start / Stop push buttons	Operator commands.
S1 photoelectric sensor	Detects package at conveyor entry.
S2 proximity sensor	Confirms package position.
S3 end/limit sensor	Detects package at final position.
Motor starter / VFD	Controls conveyor motor.
Diverter actuator	Optional sorting mechanism.
HMI	Optional status, counts and alarms.

5. BLOCK DIAGRAM

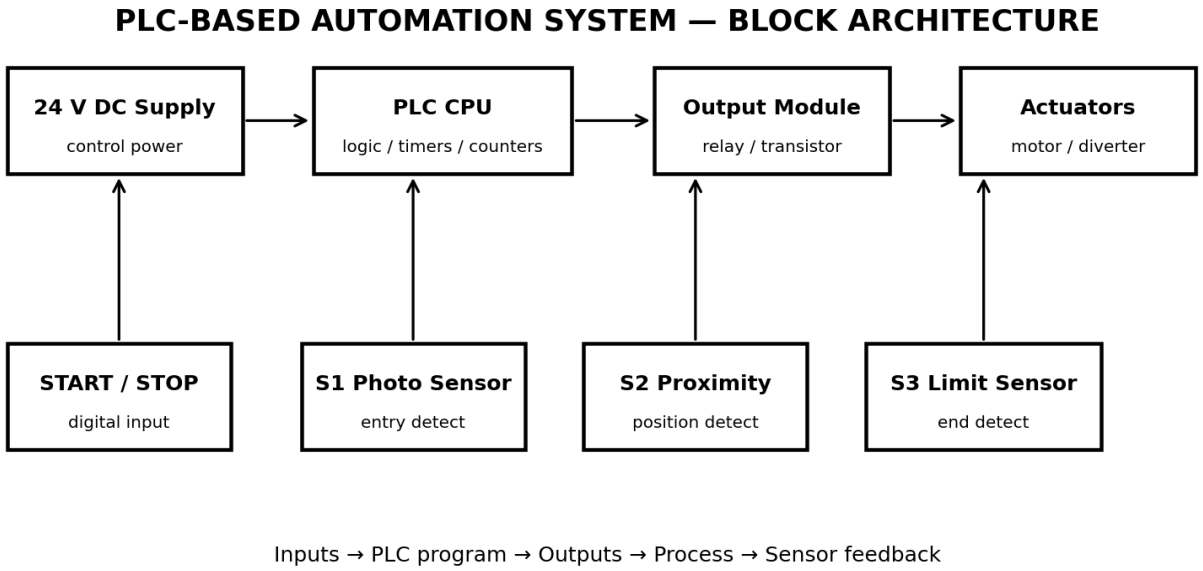


Figure 1. Proposed PLC-based automation architecture.

A commercial PLC family such as Siemens S7-1200 can be engineered and programmed with STEP 7 in TIA Portal. Siemens describes STEP 7 as an integrated environment for configuring, programming and maintaining automation systems and PLCs. [2] The same architecture can be implemented with PLCs from other manufacturers.

6. I/O ALLOCATION

Address	Tag	Type	Description
I0.0	START	DI	Start automatic mode
I0.1	STOP	DI	Normal stop command
I0.2	S1_ENTRY	DI	Package detected
I0.3	S2_POSITION	DI	Position confirmed
I0.4	S3_END	DI	End position reached
I0.5	RESET	DI	Reset counter/fault
Q0.0	MOTOR_RUN	DO	Conveyor motor command
Q0.1	DIVERTER	DO	Sorting actuator
Q0.2	RUN_LAMP	DO	Running indication
Q0.3	FAULT_LAMP	DO	Fault indication

7. CONTROL SEQUENCE

1 — Idle: MOTOR_RUN is OFF. The PLC waits for START and checks safety/permissive conditions.

2 — Run: START latches automatic mode and MOTOR_RUN turns ON.

3 — Detect: S1 and S2 confirm that a package is present and at the expected position. A timer can reject an abnormal sensor pulse.

4 — Sort: If the package is classified for rejection, the PLC energizes DIVERTER for a defined time. Otherwise, it continues along the conveyor.

5 — End: S3 detects arrival, increments the item counter and either stops the conveyor or releases the next cycle depending on the selected operating mode.

6 — Fault: A sensor timeout, overload feedback or other configured fault removes the normal run command and activates the fault indication.

8. PLC SCAN AND PROGRAMMING

A PLC normally performs a repeated scan consisting of input acquisition, program execution and output update. This cyclic behavior is well suited to conveyor control because the process is naturally sequential. Ladder Diagram is particularly suitable for teaching relay-style logic, while Structured Text or Function Block Diagram can simplify more complex calculations and reusable function blocks. IEC 61131-3:2025 provides the current standard framework for these programming languages. [1]

9. LADDER-LOGIC DESIGN CONCEPT

A simplified motor-control rung can be represented as a seal-in circuit: START sets the run state, while STOP, fault and safety conditions interrupt it. In software terms, the essential rule is that a hazardous or abnormal condition must dominate the normal run command.

Pseudocode:

```
IF NOT Safety_OK OR STOP THEN MOTOR_RUN := FALSE;
ELSE IF START THEN AUTO_MODE := TRUE;
IF AUTO_MODE AND S1_ENTRY THEN MOTOR_RUN := TRUE;
IF rising_edge(S3_END) THEN COUNT := COUNT + 1;
IF SORT_REQUIRED THEN DIVERTER := TRUE for T_SORT;
IF FAULT THEN MOTOR_RUN := FALSE; FAULT_LAMP := TRUE;
```

Timers are useful for controlling diverter pulse duration and for detecting missing sensor transitions. Counters are useful for total production, accepted items and rejected items. One-shot/rising-edge logic prevents a continuously active S3 sensor from incrementing the counter on every PLC scan.

10. HMI AND INDUSTRIAL COMMUNICATION

An HMI can show Auto/Manual mode, motor state, sensor states, package count, rejected count, alarms and reset controls. A more advanced design can connect the PLC to supervisory software or manufacturing systems through an appropriate industrial network. Production data such as cycle time, motor run time and fault history can then be collected for analysis.

The proposed architecture is compatible with an Industry 4.0 development path. The basic cell can first be validated locally, then connected to higher-level systems after the control sequence is stable. Connectivity should be designed so that normal IT/data functions do not compromise safety-related functions.

11. SAFETY DESIGN

Machine safety requires more than ordinary PLC logic. ISO 13849-1:2023 provides a methodology and requirements for designing and integrating safety-related parts of control systems. [3] For an actual conveyor, emergency-stop functions, guarding, access control, motor isolation, overload protection and other safeguards should be selected using a machine risk assessment.

The standard PLC program should not automatically be considered a safety controller. Where a safety function requires a safety-rated architecture, appropriate safety components and validation must be used. For a student prototype, low-voltage actuators, physical guarding and supervised operation are recommended.

12. TESTING AND VALIDATION PLAN

Testing should progress from individual I/O checks to complete sequence testing. First verify each push button and sensor. Then test the motor command without a load. Next introduce packages and confirm detection, transport, sorting and counting. Finally test abnormal conditions and recovery.

Test	Expected result
START pressed	Automatic mode becomes active; motor runs if safe.
STOP pressed	Motor command turns OFF.
S1 detects package	PLC recognizes package entry.
S3 detects package	Counter increments once per package.
Sorting condition true	Diverter activates for configured time.
Sensor timeout	Fault state prevents normal operation.
Emergency stop	Hazardous motion is removed through the designed safety function.
RESET after fault cleared	Fault indication clears and system can restart safely.

13. EXPECTED RESULTS

The prototype should transport packages consistently, detect their presence, perform the defined sorting decision and maintain a reliable completed-item count. It should also respond predictably to Stop, Reset and configured fault conditions. The expected educational outcome is an understanding of how a PLC converts sensor information into deterministic actuator commands.

14. ADVANTAGES

PLC-based automation provides flexible software control, repeatable operation, easy integration of sensors, simple implementation of timers and counters, and straightforward expansion. A single PLC can coordinate multiple stations and can exchange data with HMI and supervisory systems.

15. LIMITATIONS

The system requires correct electrical integration, programming and commissioning. Complex motion may require servo or motion-control hardware. Unstructured objects may require machine vision. PLC control logic alone is not a substitute for an engineered machine-safety system. The proposed project is a conceptual educational design and not a certified production machine.

16. FUTURE SCOPE

Future work can add barcode/RFID identification, machine vision, VFD speed control, servo positioning, automatic recipe management, energy monitoring, remote diagnostics and production dashboards. A digital twin or simulation environment could be used to validate the sequence before commissioning. Predictive maintenance could use motor current, vibration and cycle-time data to identify developing faults.

17. IMPLEMENTATION PLAN

Phase 1 — Requirements: Define package size, throughput, sensor locations, actuator requirements and safety constraints.

Phase 2 — Electrical design: Select PLC I/O, power supply, sensors, motor interface and operator devices; create the wiring diagram.

Phase 3 — Software: Implement the state sequence, interlocks, timers, counters, fault handling and HMI tags.

Phase 4 — Commissioning: Test I/O, run the sequence without load, introduce packages and verify abnormal cases.

Phase 5 — Optimization: Measure cycle time and reliability, tune sensor delays and diverter timing, and add production data collection.

18. CONCLUSION

This mini project demonstrates a complete PLC-based automation concept for conveyor sorting and counting. Sensors provide real-time process information; the PLC executes deterministic logic; and outputs control the motor and optional sorting mechanism. Timers, counters, interlocks and fault handling make the design suitable for a realistic automation laboratory project. With appropriate safety engineering and validation, the same architecture can be adapted to packaging, assembly, inspection and material-handling applications.

REFERENCES

[1] International Electrotechnical Commission (IEC). (2025). *IEC 61131-3:2025 — Programmable controllers — Part 3: Programming languages*. IEC, Geneva. https://webstore.iec.ch/en/publication/68533

[2] Siemens. (2026). *STEP 7 TIA Portal – PLC Programming Software*. Siemens Digital Industries. https://www.siemens.com/en-us/products/tia-portal/step7/

[3] International Organization for Standardization (ISO). (2023). *ISO 13849-1:2023 — Safety of machinery — Safety-related parts of control systems — Part 1: General principles for design*. ISO, Geneva. https://www.iso.org/standard/73481.html

[4] Siemens. (2022). *S7-1200 Programmable Controller System Manual, V4.6*. Siemens Industry Online Support.

[5] Bolton, W. (2021). *Programmable Logic Controllers*, 7th ed. Elsevier.

Note: This report is an educational proposal. A real machine requires detailed electrical, mechanical and functional-safety engineering, component ratings, risk assessment and commissioning by qualified personnel.